Solution Requirements

Date	4 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	4 Marks

Solution Requirements

Project Overview

The **Flood Prediction System using Machine Learning** is designed to predict the likelihood of flood events using historical and real-time weather parameters such as annual rainfall, cloud cover, humidity, temperature, and seasonal rainfall. The system employs machine learning algorithms including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and XGBoost to classify flood risk accurately.

The application provides an intuitive web interface developed using the Flask framework, enabling meteorologists and disaster management authorities to input weather parameters and receive instant flood risk predictions. The solution supports early warning, disaster preparedness, and efficient resource allocation.

Purpose of Solution Requirements

This document defines the functional and non-functional requirements of the Flood Prediction System. It specifies the system's expected functionalities, performance characteristics, security requirements, usability standards, and operational constraints to ensure reliable and accurate flood prediction.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)

1	Authentication	Users securely log in using a username and password to access the Flood Prediction System.	High
2	Authorization levels	Admins manage datasets and machine learning models, while authorized users can submit weather data and view predictions.	High
3	External interfaces	The system integrates with historical weather datasets and can connect to weather APIs for real-time weather information.	Medium
4	Transactions processing	Weather parameters are processed through machine learning models to generate flood risk predictions instantly.	High
5	Reporting	The system provides prediction reports, model accuracy, historical records, and dashboard visualizations.	Medium
6	Business rules, etc.	Flood prediction is performed only when all mandatory weather parameters are entered. Invalid or incomplete inputs are rejected.	High

7	Compliance to laws or regulations	The application follows data privacy principles and ethical AI practices while handling user and weather data.	Medium
8	Other	Administrators can retrain and update the prediction model using newly available weather datasets.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	**NFR Category**	**Requirement Description**	**Target Metric / Acceptance Criteria**
1	Performance & Speed	The system should provide flood predictions with minimal delay.	Prediction generated within **2 seconds**
2	Scalability	The application should support increasing users and larger weather datasets without performance degradation.	Supports **500+ concurrent users**
3	Security & Data Privacy	User credentials and application data should be protected through secure authentication and encrypted communication.	Secure login and HTTPS communication
4	Reliability & Availability	The system should remain operational during emergencies with high availability.	99.9% uptime

5	Usability & Accessibility	The application should provide a simple, responsive, and user-friendly interface across different devices.	Responsive design with intuitive navigation
6	*Other*	The software should be modular, maintainable, and easy to update with improved machine learning models.	Modular architecture supporting future enhancements